CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 11: Areas Related to Circles

Previous year board questions, topic-wise: 10 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Sector and segment of a circle

1. A sector of a circle of radius 21 cm has a central angle of 60°. Find the length of its arc. (a) 22 cm (b) 44 cm (c) 88 cm (d) 11 cm CBSE 2026, 1 mark

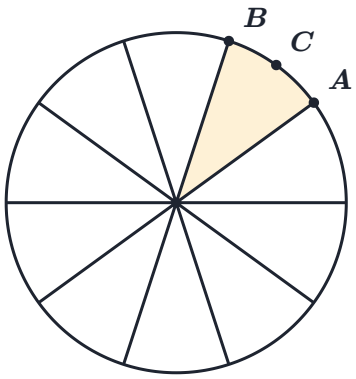
2. The hour hand of a clock is 7 cm long. Through what angle does it turn between 7:00 a.m. and 8:10 a.m.? (a) $\left(\frac{35}{4}\right)^\circ$ (b) $\left(\frac{35}{2}\right)^\circ$ (c) 35° (d) 70° CBSE 2026, 1 mark

3. A circular ornament of diameter 35 cm is made from silver wire. Wire is also used for 5 diameters, which cut the circle into 10 equal sectors. (i) Find the radius. (ii) Find the circumference. (iii) Find the total length of wire needed, OR the area of each sector. CBSE 2026, 4 marks

4. A sector with a central angle of 72° has an area of 40π square units. Find the radius of the circle. CBSE 2025, 1 mark

5. A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? CBSE 2025, 1 mark

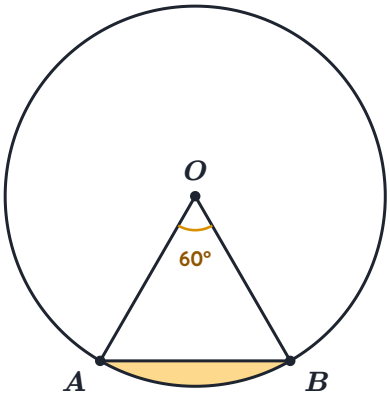
6. A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. A and B are the ends of one sector on the circle and C lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc ACB. (iii) Find the area of each sector, OR the total length of wire used. CBSE 2025, 4 marks



7. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. CBSE 2024, 5 marks

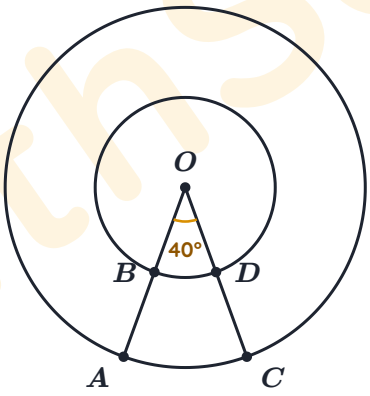
8. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the minor segment and the area of the major segment. Take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$.

CBSE 2023, 5 marks

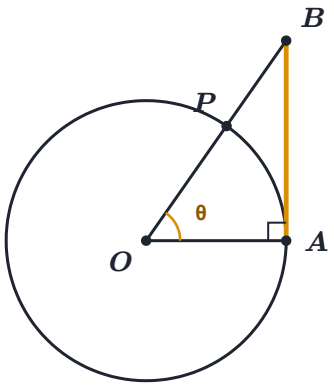


9. Two concentric circles with centre O have radii 7 cm and 14 cm. A sector of angle 40° is cut away from the ring between them. Find the area of the part of the ring that remains. Take $\pi = \frac{22}{7}$.

CBSE 2016, 3 marks



10. OAP is a sector of a circle with centre O, radius r and angle θ (in degrees). AB is perpendicular to the radius OA and meets OP produced at B. Prove that the perimeter of the region bounded by AB, BP and the arc PA is $r\left[\tan\theta + \sec\theta + \frac{\pi\theta}{180} - 1\right]$. CBSE 2016, 4 marks



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